Evolve Together, Fall into Place: How Social Scaffolding Shapes Self-Evolving Multi-Agent Systems

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HarnessRSI 65 ★★

evidence: retained

Paper

1 What changes, what stays

Weights fixed. Skills/memory are role-local; the constitution is team-wide. No pooled skill library is described. None skips L2, not task messaging.

Composition: all weak / mixed / all strong
Rules: none / collaboration / conformity / competition (permissions, sharing, decisions)

Task session
Chair delegates; task messages are shared.
Training outcomes and traces feed learning.

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L1 · Retain individual changes
SkillOpt skills: validation improvement gate.
Meta-Team/MemRL: no L1 gate.

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L2 · Learn about teammates
Notes enter later prompts and update models.
Tier 1 reads all notes; Tier 2 only its own.

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L3 · Propose team changes
Revise roles, prompts, configs, constitution.
Who proposes/reviews/decides depends on rules.

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L3 gate · Block clear harm
Same 20 validation tasks and seeds per side.
Revert clear loss; otherwise keep (α = 0.3).

Competition then promotes the top scorer.
Promotion is ungated; freeze team for test.

2 Selected gains, not a universal rule

SkillOpt · test success (%) · evolved strong solo
Weak: Qwen3-30B-A3B-Instruct-2507
Strong: Qwen3.6-35B-A3B · thinking off · T = 0.1

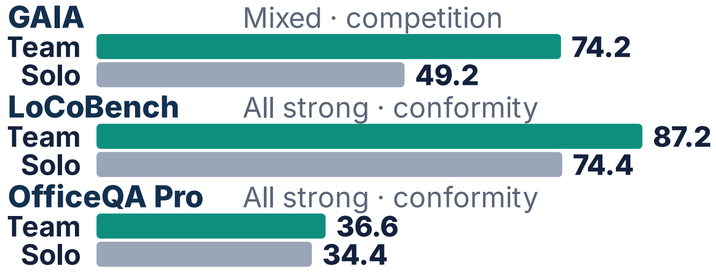


Table 4 means; SD (team / solo): GAIA 3.8 / 1.4; LoCoBench 2.1 / 2.5; OfficeQA 6.5 / 0.9. Three frozen-team evaluations, one evolution run per condition. Best team/structure chosen post hoc.

3 Controls & costs that matter

- **Controls:** preevolve has zero updates; none keeps L1/L3 and the same L3 gate, without L2 notes.
- **Both solos:** all benchmark tools, same L1; no constitution/L2/L3. SkillOpt/MemRL retain the strong updater.
- **Signal:** SkillOpt sees training ground truth; MemRL gets success/failure. L2 observes peers.
- **Selection:** 20 val tasks; tests 40 / 99 / 62. MemRL thresholds: unlabeled val–train inputs. Other default-selection splits/tuning budgets incompletely specified.
- **Statistics:** 24/27 best structures beat none; only 7 paired $p < 0.05$. Authors: optimistic p-values from post-hoc selection and correlated repeats.
- **Partial budget:** 900 s/session; 100 team messages vs 50 solo; 50 steps/agent. Reflection and validation cost extra.
- **Denominator:** time/message-limit terminations count as failures, including in gates; crashed sessions are excluded.
- **Deployment ≠ search:** GAIA mixed competition: 19.4 calls/session vs 46.8 preevolve; not optimization cost (Table 5).
- **Whole-experiment time:** GAIA SkillOpt 4.5–16.0 h; LoCoBench 14.5–37.2 h per GPU pair. No optimizer-only token/$ or full cost-matched solo table.
- **Leakage:** disjoint splits stated; web unrestricted, no empirical contamination audit. No observed leakage is claimed.

Authors' caveat: shared-GPU load plus time limits may favor fewer active agents.

4 Where gains fail or are thin

- **OfficeQA mixed:** best 21.5 ± 2.4, below unevolved strong solo 28.5 ± 0.9. All-strong 36.6 vs 34.4 is inside evaluation noise.
- **Evolution alone can work:** GAIA all-strong SkillOpt none 60.8 ± 5.8; best social row 57.5 ± 6.6.
- **Competition can collapse:** LoCoBench all-weak Meta-Team none 43.4 → competition 22.6.
- **Not universal across families:** Mistral LoCoBench SkillOpt all-strong best social 58.9 ± 4.7 vs evolved solo 80.5 ± 3.5 (Table 27).

Takeaway

A team can grow a better retained harness, then need fewer teammates. The large gains cluster on GAIA and LoCoBench; who leads matters, but no social structure reliably wins everywhere.

Our read · HarnessRSI

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Test + social controls; evaluation SD only.
Not established: evolution robustness, artifact transfer, full cost matching, leakage audit.